xLSTM RUL

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ABSTRACT

bla bla bla

1 Introduction

- nasa pyprog library: https://nasa.github.io/progpy/ - github? - synthetic data - https://github.com/thummd/CausalTimePrior - https://ch.mathworks.com/de/company/technical-articles/three-ways-to-estimate-remaining-useful-life-for-predictive-maintenance.html - https://data.phmsociety.org/nasa/ - mehrere prior oder nicht

- felix: eval benchmark datensätze ruassuchen. Univariat?

2 Task description and data construction

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Task modeling. We approach this task as a regression problem. For every item and shop pair, we need to predict its next month sales(a number).

2.1 Headings: second level

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$$\xi_{ij}(t) = P(x_t = i, x_{t+1} = j | y, v, w; \theta) = \frac{\alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})}{\sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})} \tag{1}$$

2.1.1 Headings: third level

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3 ML Modelling Steps

- Tirex: [1] [2]
- Tirex is easy to run and can be used on consumer hardware: `https://github.com/NX-AI/tirex/blob/main/examples/quick_start_tirex.ipynb`
- Chronos-2 has a similar API and could be called easily: `https://github.com/amazon-science/chronos-forecasting`
- As Chronos has quite a bunch of different models, we should compare chronos-2-small against Tirex, as both have a comparable number of parameters.
- Gathering tutorials: `https://github.com/pauly-andreas/tirex-test`
- TSFM for RUL [3]

4 Examples of citations, figures, tables, references

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The documentation for `natbib` may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

> Hasselmo, et al. (1995) investigated...

`https://www.ctan.org/pkg/booktabs`

4.1 Figures

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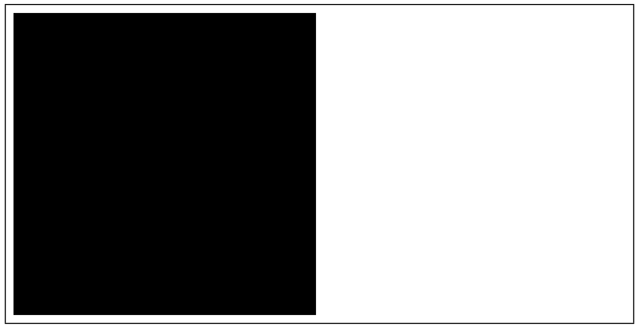

Figure 1: Sample figure caption.

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0	! ВО ВЛАСТИ НАВАЖДЕНИЯ (ПЛАСТ.) D	0	40
1	!ABBYY FineReader 12 Professional Edition Full...	1	76
2	***В ЛУЧАХ СЛАВЫ (UNV) D	2	40
3	***ГОЛУБАЯ ВОЛНА (Univ) D	3	40
4	***КОРОБКА (СТЕКЛО) D	4	40

add footnotes. [1] Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetuer eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor.

4.2 Tables

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4.3 Lists

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- consectetur adipiscing elit.
- Aliquam dignissim blandit est, in dictum tortor gravida eget. In ac rutrum magna.

[1] Sample of the first footnote.

Table 1: Sample table title

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

References

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[3] Amir El-Ghoussani, Michele De Vita, Ronald Naumann, and Vasileios Belagiannis. Time-series foundation model embeddings for remaining useful life estimation, 2026.

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[6] Guy Hadash, Einat Kermany, Boaz Carmeli, Ofer Lavi, George Kour, and Alon Jacovi. Estimate and replace: A novel approach to integrating deep neural networks with existing applications. *arXiv preprint arXiv:1804.09028*, 2018.